

HOW DO YOU IMPLEMENT HEIJUNKA IN YOUR ORGANIZATION?



IMPLEMENT HEIJUNKA IN YOUR ORGANIZATION

HEIJUNKA DOES NOT BUILD PRODUCTS ACCORDING TO THE ACTUAL FLOW OF CUSTOMER ORDERS

TAKES THE TOTAL VOLUME OF ORDERS BEING MADE IN A SPECIFIC PERIOD AND LEVELS THEM OUT



IMPLEMENT HEIJUNKA IN YOUR ORGANIZATION



iPhone XR



iPhone XS



iPhone XS Max

EXAMPLE – TRADITIONAL PRODUCTION (UNLEVELED)

Monday
Production



iPhone XS



iPhone XS



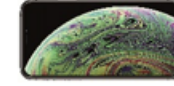
iPhone XS



iPhone XS



iPhone XS



iPhone XS



iPhone XS



iPhone XS

Tuesday
Production



iPhone XS



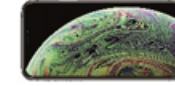
iPhone XS



iPhone XS



iPhone XS



iPhone XS



iPhone XS



iPhone XS



iPhone XS

Wednesday
Production



iPhone XS



iPhone XS



iPhone XS



iPhone XS

Changeover



iPhone XR



iPhone XR

Thursday
Production



iPhone XR



iPhone XR



iPhone XR



iPhone XR



iPhone XR



iPhone XR



iPhone XR

Friday
Production



iPhone XR

Changeover



iPhone XS Max



iPhone XS Max



iPhone XS Max



iPhone XS Max

ISSUES WITH UNLEVELED SCHEDULE

CUSTOMER BUYING PATTERN IS UNPREDICTABLE

THERE IS A RISK OF UNSOLD GOODS

THE USE OF RESOURCES IS UNBALANCED

IT PLACES AN UNEVEN DEMAND ON UPSTREAM PROCESSES

BULLWHIP EFFECT – MULTIPLIES BEHAVIOR BACKWARDS THROUGH THE SUPPLY CHAIN | A SMALL CHANGE IN IN THE SCHEDULE WILL RESULT IN EVER-INCREASING INVENTORY BANKS AT EACH STAGE OF THE SUPPLY CHAIN

EXAMPLE – TRADITIONAL PRODUCTION (UNLEVELED)

Monday
Production



iPhone XS



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Wednesday
Production



iPhone XS



iPhone XS



iPhone XS



iPhone XS

Changeover



iPhone XR



iPhone XR

Thursday
Production



iPhone XR



iPhone XR



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iPhone XR



iPhone XR



iPhone XR

Friday
Production



iPhone XR

Changeover



iPhone XS Max



iPhone XS Max



iPhone XS Max



iPhone XS Max

ROOT-CAUSES & SOLUTIONS

POTENTIAL REASONS FOR LONG CHANGEOVER TIME:

**DIFFERENT PARTS REQUIRED FOR
DIFFERENT PHONE VARIANTS**

**DIFFERENT SIZED INSTRUMENTS
TO ASSEMBLE EACH VARIANT**

**EMPLOYEES ARE SKILLED
DIFFERENTLY TO MANUFACTURE
DIFFERENT VARIANTS**

SOLUTIONS IDENTIFIED AFTER ROOT-CAUSE ANALYSIS:

**REDESIGNED WORKSTATION TO
HOUSE DIFFERENT PARTS**

**ALSO REDESIGNED TO HOUSE
DIFFERENT SIZED INSTRUMENTS**

**CROSS-SKILLED EMPLOYEES TO
MANUFACTURE DIFFERENT
VARIANTS OF THE PHONE**

EXAMPLE – MIXED MODEL ASSEMBLY LINE

Monday
Production



iPhone XS



iPhone XS



iPhone XR



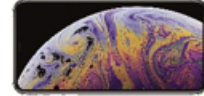
iPhone XS



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iPhone XS Max



iPhone XR

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iPhone XS



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iPhone XS Max



iPhone XR

Wednesday
Production



iPhone XS



iPhone XS



iPhone XR



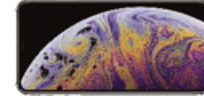
iPhone XS



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iPhone XS Max



iPhone XR

Thursday
Production



iPhone XS



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iPhone XR

Friday
Production



iPhone XS



iPhone XS



iPhone XR



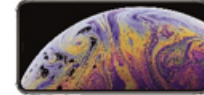
iPhone XS



iPhone XS



iPhone XS



iPhone XS Max



iPhone XR

BENEFITS OF LEVELING THE SCHEDULE

FLEXIBILITY TO MAKE WHAT THE CUSTOMER WANTS WHEN THEY WANT IT

REDUCED RISK OF UNSOLD GOODS

BALANCED USE OF LABOR AND MACHINES

SMOOTHED DEMAND ON UPSTREAM PROCESSES AND THE PLANT'S SUPPLIERS

THE PLANT WAS ABLE TO ELIMINATE THE SETUP TIME FOR CHANGEOVER

STEP 01: IDENTIFY

IDENTIFY THE PRODUCTS THAT NEED TO BE PRODUCED

IDENTIFY THEIR MONTHLY/SEASONAL PRODUCTION REQUIREMENTS

IDENTIFY THE WORKSTATION SET UP REQUIREMENTS

IDENTIFY THE REQUIREMENTS OF INSTRUMENTS REQUIRED FOR DIFFERENT PRODUCT VARIANTS

IDENTIFY THE EMPLOYEE SKILL-SET REQUIREMENT

STEP 02: MODIFY

MAKE CHANGES TO THE WORKSTATION SET UP

MAKE CHANGES TO THE PLACEMENT OF INSTRUMENTS

CROSS-SKILL EMPLOYEES BY PROVIDING THEM REQUIRED TRAINING

CREATE A MIXED MODEL SCHEDULE FOR PRODUCTION

STEP 03: IMPLEMENT

IMPLEMENT A PILOT

SEEK FEEDBACK FROM THE PILOT TEAM

MAKE NECESSARY CHANGES TO THE SET UP AND SCHEDULE

FINALLY, IMPLEMENT THE NEW LEVELLED WORKLOAD SCHEDULE